Agenda- Group Project

Build a consistent implied monetary policy pricer for global EM rates (and ideally developed markets too), replacing reliance on Bloomberg MIPR. The intern is expected to: (1) work with cross-regional sponsors to agree the markets in scope (starting with Latam, CE3, Turkey, and extending to Asia and Africa); (2) for each market, identify the correct instruments to use (swaps, FRA, bonds, or a combination) and the right interpolation methods; (3) build a pricer that shows implied pricing per central bank meeting date (not just fixed horizons); (4) expose the underlying assumptions so users can change them; and (5) as an extension, allow users to enter specific central bank policy path scenarios and compute “fair value” against them. Users span research, sales, and trading, so the tool should be accessible to all three.

91) World Map | 92) Refresh | Market Implied Policy Rates

Regional | Analyze 1Y | Range | Historical

Country	Policy Rate	Policy Efctv	Policy Basis	Policy Meeting	Implied Policy 1M	3M	6M	1Y	2Y	3Y	5Y	Curve	Total Change 1Y	1Y Over 30 Days Low	Range	High
1) Americas																
10) United States	3.63	3.630	0.5	07/29	3.66	3.83	4.07	4.15	3.88	3.82	3.91		53	3.86		4.18
11) Canada	2.25	2.300	5.0	07/15	2.23	2.25	2.37	2.68	2.96	2.93	3.07		43	2.66		2.88
12) Mexico	6.50	6.500	0.0	06/25	6.61	6.68	6.79	7.25	7.92	8.17	8.40		75	6.99		7.61
13) Chile	4.50	4.500	0.0	07/28	4.49	4.53	4.49	4.44	4.74	4.89	5.14		-6	4.34		4.86
14) Brazil	14.25	14.150	-10.0	08/05	14.25	14.23	14.68	15.50	15.32	15.16	14.88		125	13.89		15.66
2) EMEA																
20) Eurozone	2.25	2.181	-6.9	07/23	2.27	2.40	2.57	2.65	2.54	2.60	2.78		40	2.59		3.05
21) United Kingdom	3.75	3.729	-2.1	07/30	3.76	3.84	4.02	4.17	4.04	4.07	4.28		42	4.09		4.37
22) Switzerland	0.00	0.000	0.0	09/24	-0.05	-0.04	0.04	0.11	0.23	0.37	0.61		11	0.04		0.30
23) Norway	4.25	4.570	32.0	08/13	4.19	4.34	4.57	4.49	3.99	3.79	3.50		24	4.34		4.57
24) Sweden	1.75	1.748	-0.2	08/20	1.75	1.81	1.97	2.18	2.34	2.46	2.72		43	2.07		2.38
25) Denmark	1.85	1.724	-12.6		2.14	2.27	2.50	2.63	2.57	2.64	2.81		78	2.28		2.66
26) Czech Republic	3.75	3.310	-44.0	08/06	3.86	3.86	4.06	4.15	4.32	4.24	4.30		40	3.93		5.27
27) Poland	3.75	3.840	9.0	07/08	3.51	3.78	3.77	4.17	3.92	3.93	4.16		42	4.07		4.78
3) Asia/Pacific																
30) Australia	4.35	4.350	0.0	08/11	4.35	4.43	4.51	4.43	4.17	4.22	4.53		8	4.40		4.63
31) New Zealand	2.25	2.250	0.0	07/08	2.40	2.61	2.85	3.20	3.47	3.61	3.96		95	3.12		3.62
32) Japan	1.00	0.977	-2.3	07/31	1.00	1.03	1.20	1.48	1.92	2.20	2.67		48	1.31		1.66
33) China	1.40	1.480	8.0		1.41	1.37	1.37	1.35	1.40	1.47	1.61		-5	1.33		1.50
34) India	5.25	5.380	13.0	08/05	5.24	5.25	5.56	6.04	6.09	6.27	6.44		79	5.98		6.36
35) Korea	2.50	2.920	42.0	07/16	2.40	2.77	3.25	3.69	3.70	3.69	3.70		119	3.48		3.93

Expose the underlying assumptions so users can change them:

- Ex: Right now MIPR just shows Brazil 1y implied rate = x%. but you don’t know which instruments were used, which interpolation, which day count, whether policy changes are assumed only at meetings, how the first fixing is handled

<u>Main purpose of this tool --- allow users to express and test their views</u>

Questions to ask:

- Could you walk us through a real example of how you would use this tool? Suppose you had a strong view on X or Y country, what workflow do you imagine

1.) Agree on markets in scope (cross regional sponsors)
    - Who are these regional sponsors
    - Does each currency use multiple instruments or should we research which instrument is most common for each EM?

        - Is it based on what instruments are most liquid in that market?
        - Should users be able to choose btw swaps, bonds, futures as the curve course

2.) Build out
    - Can we download central bank meeting dates from Bloomberg?
    - How far in advance do they want to see implied rates (1 year 5 year)?
    - Rate hike or rate cut? For underlying assumptions (GDP, unemployment, inflation, labour data) (Hawkish or dovish)
    - What are these policy path scenarios?
    - Where do we derive fair value?
        - Given your policy path, what should the market instrument be worth?

Background Info to Know:

- Swaps
- FRA: forward rate agreement
- Bonds

What goes into **predicting global interest rates**? - combination of macro conditions and monetary policy decisions

- Analyzing macro data
    - CPI
    - Personal consumption expenditures
    - Labor markets- unemployment, new jobs
- Tracking central bank policies
- Read the bond market (yield curve)
    - Inverted curve= short-term rates are higher than long term rates

Aggregate Expert Forecasting Models:

- Track economic calendars (trading economics calendar)
- OECD Projections (long-term government bond yields)

Underlying assumptions built-in:

- Hawkish vs. Dovish expectation
- Expected inflation changes
- GDP change
- Currency appreciation vs. depreciation

Latam:

- Mexico
- Brazil
- Chile

- Colombia

CE3:

Turkey:

Asia:

- India
- Indonesia (economic data [Indonesia | Data])
- Thailand
- Malaysia
- Philippines

Africa:

Core Economic Indicators to build into our model:

- GDP
- Unemployment rate
- Inflation rate
- Interest rates
- Trade balance (difference between exports and imports) (surplus vs deficit)
- Government debt to GDP

FOUND at [International Monetary Fund | IMF]

- Can pull an excel file on data for these countries

Notes from meeting with Gordian

People to connect with:

- Arup Ghosh
- Marcos Ang
  - Asian CB pricers
  - Singapore
- Ethan Lester
  - London

**Check the loan market compendium on website

- Tool that can derive from instruments what market prices in for each CB meeting, derived from the curve
- Concept of smooth curve, work w/ discount factors, weight of future payment in todays terms
- Concavity – can't do linear interpolation because it underestimates
    - Exponential spline
- Bootstrapping of a curve

What Gordian sees as the UI of this tool:

- Wants to be able to "reverse it"
- Trying to solve what is market implied fair value
- Wants "what if" scenario analysis
- Ex: You are long X, you think CB will hike, what is your risk
    - How will 1 yr change if you change the scenario
- Sheet where you can update the what if
    - Use WIRP Chile as example

Showing what the interest rate market is pricing for Chile's CB policy path using Overnight Index Swaps (OIS)

- OIS is a derivative where to parties exchange:
    - Fixed rate (what the market implies)
    - Floating rate (the realized overnight rate, compounded over time)

#Hikes/Cuts (Expressed in number of 25bp moves" priced in); +1.0 = 1 full hike (25bp)

- -0.282 --> means ~28% probability of a 25 bp cut
- +0.200 --> ~20% chance of a 25bp hike
- -0.664 --> ~66% chance of a 25 bp cut

%Hike/Cut (Expressed as a percentage probability)

Implied Rate

- The market-implied policy rate after that meeting

Enable Overrides

Region: Chile » Instrument: Overnight Index Swaps »

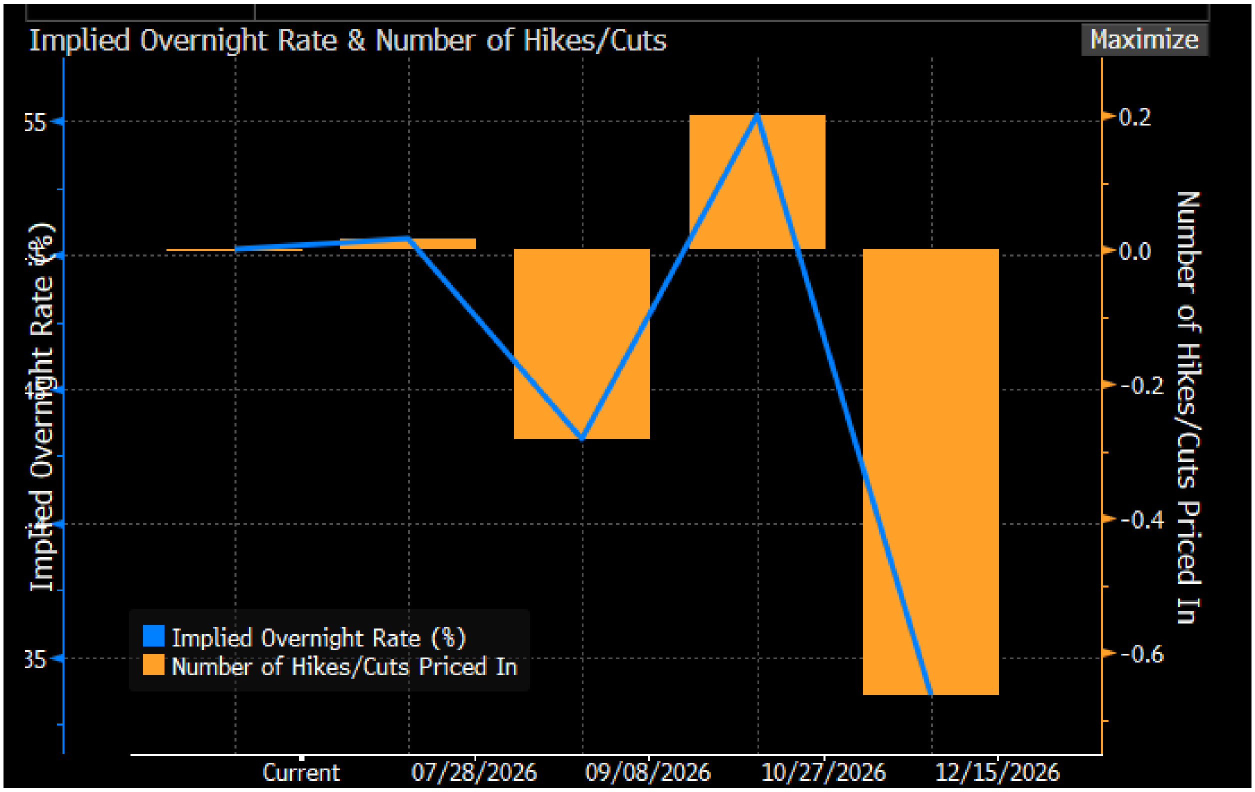
Target Rate 4.50 Pricing Date 06/23/2026

Effective Rate 4.50 Cur. Imp. O/N Rate 4.502

Meeting	#Hikes/Cuts	%Hike/Cut	Imp. Rate Δ	Implied Rate	A.R.M.
07/28/2026	+0.016	+1.6%	+0.004	4.506	0.250
09/08/2026	-0.282	-29.8%	-0.071	4.431	0.250
10/27/2026	+0.200	+48.2%	+0.050	4.552	0.250
12/15/2026	-0.664	-86.4%	-0.166	4.336	0.250

- Blue line = implied overnight rate (shows expected path over time)
- Orange bars = number of hikes/cuts priced
  - Negative bars = cuts
  - Positive bars = hikes
- Market expects easing overall, but with some uncertainty and volatility in the path
  - Near term (july) -- market expects no change
  - September (leaning toward a rate cut, ~30% priced)
  - October (some chance of a hike, suggesting uncertainty/data dependence)
  - December (Strong expectation of cuts, ~66% probability of easing)

INTERNAL



CLP OIS curve builder

- Cash rate (anchor)
    - 1 Day: 4.50 / 4.50
        - Current overnight policy rate
        - Starting point of the entire curve
- No Futures / FRAs
    - Chile doesn't have liquid short-term futures
    - So the curve relies almost entirely on OIS swaps
- Swap rates (core inputs)
    - OIS swap rates from 1 month to 20 years
        - These are the fixed rates at which the market is willing to swap versus overnight CLP
        - They embed expectations of future short-term rates


CLP 193 - CLP (vs. Camara) Name CLP (vs. Camara) Default Privilege Global 06/23/26
Curve Construction Curve Analysis
Shift +0.00 bp Legend

Cash Rates PCS CMPN

Term	Bid	Ask
1 DY	4.50000	4.50000

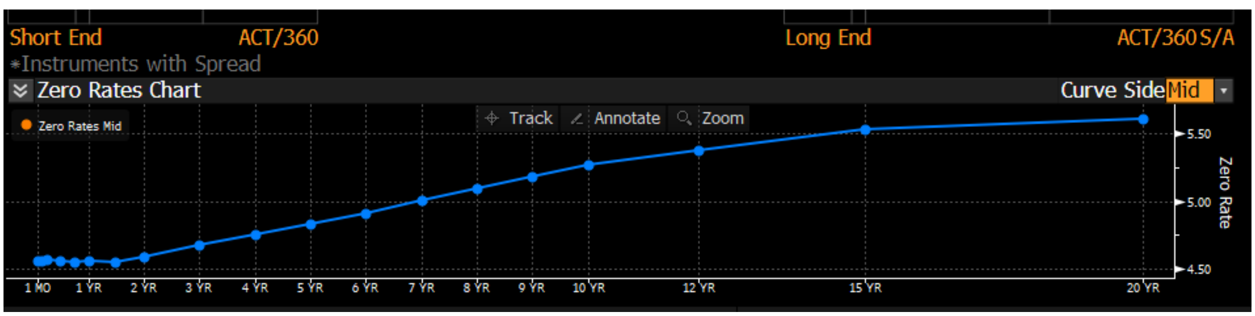
There are no Futures or FRA contracts available for this curve.

Swap Rates PCS BGN

Term	Bid	Ask
1 MO	4.48987	4.53013
2 MO	4.49509	4.53491
3 MO	4.50953	4.55047
6 MO	4.52468	4.56532
9 MO	4.54440	4.58560
1 YR	4.57949	4.62051
18 MO	4.62468	4.66532
2 YR	4.55984	4.60016
3 YR	4.64524	4.68476
4 YR	4.71532	4.75468
5 YR	4.78502	4.82498
6 YR	4.85992	4.90008
7 YR	4.94496	4.98504
8 YR	5.02491	5.06509
9 YR	5.09986	5.14014
10 YR	5.16974	5.21026
12 YR	5.26472	5.30628
15 YR	5.38969	5.43031
20 YR	5.45400	5.50100

Zero rate curve

- Shows zero rates (spot rates) derived from those swaps
- Upward sloping curve --> the market expects higher average rates in the long run than today
-



WIRP

- US: Fut, OIS <-- try with this
- EM will be swap based
- Try to replicate Chile --> understand Colombia
    - Chile: if 3month, but no 1m, 2m --> have to make an assumption. Extrapolate to get 1m from 3m
- **He is interested in at least 1yr dates ahead

MIPR

- US is a benchmark

- ***Want for LATAM, Mexico, Chile
    - Mexico has funky definitions
    - 28 days month (Aztec calendar)
    - TIIE = Interbank swap rate, based on 28 days
    - 1yr = 364 days
    - 10 yr is missing out on 12 days

***Need to factor in basis

MXSW

MPSW

- F10J MXMOIS 130 x 1
    - 10 yrs, 28 days

CG --- gives the curve

- Green = current
- 5583 = name of curve
- Mid YTM
- FWCM – fwd curve matrix
    - Curve list
        - Starts with 1 yr, need to find 1m

INTERNAL


MXN OIS 130x1 Curncy
GC
Related Functions Menu
Message
MXN TIIE-F RFR Curve
Custom Curve
Actions
Table
Export
Settings
Graph Curve
X-Axis Tenor
Y-Axis Mid YTM
Currency None
PCS BGN
Lower Chart Table
Specific MM/DD/YY
Relative Last 1D 1W 1M Modify
Curves & Relative Value
S583 MXN TIIE-F RFR Curve Last Mid YTM 16:23:05
8.00
7.50
7.00
6.50
1M 6M 2Y 4Y 7Y 10Y 15Y 20Y 30Y
Tenor
All Tenors
Key Tenors
Curve Id 3M 1Y 2Y 3Y 5Y 10Y 15Y 30Y
11) S583 6.545 6.700 7.030 7.265 7.560 8.010 8.245 8.305
Plot Curves
Add Curve
Browse | CRVF »
MXN TIIE-F RFR Curve
Mid YTM
+ Add Field
Base Curve S583 Mid YTM
Show Constituents on Base Curve
Recent Curves
Curves related to MPSWF10J Curncy
Bond Spread to Curve
Add Security
Plot New Issues / Points
Interpolate Curves

CDIE (Brazil)

80) CDI Estimate | 81) CDI Calculator

Pricing Date 06/23/26 | BZDIOVRA 14.150 | IDIX9 514193.97

	Tkr	DI Mty	Last	Est	Last - Est	COPOM Eff	COPOM Est	COPOM + CDI	Open Int	Volume	IDI Est
1)	ODN26	07/01/26	14.153	14.150	0.003			14.150	8106884	17018	515816.77
2)	ODQ26	08/03/26	14.152	14.150	0.002	08/06/26	-6.63	14.084	806078	33121	522085.07
3)	ODU26	09/01/26	14.115	14.126	-0.011	09/17/26	7.60	14.160	1247637	5684	527852.88
4)	ODV26	10/01/26	14.100	14.124	-0.024			14.160	2245627	99180	533694.82
5)	ODX26	11/03/26	14.130	14.132	-0.002	11/05/26	19.66	14.356	386224	510	539617.10
6)	ODZ26	12/01/26	14.162	14.167	-0.005	12/10/26	25.79	14.614	704251	7445	545095.22
7)	ODF27	01/04/27	14.190	14.227	-0.037	01/21/27	21.00	14.824	6967782	189048	551590.51
8)	ODG27	02/01/27	14.292	14.287	0.005			14.824	151558	401	557622.52
9)	ODH27	03/01/27	14.365	14.344	0.021	03/04/27	12.70	14.951	131430	384	563155.58
10)	ODJ27	04/01/27	14.390	14.411	-0.021	04/22/27	5.64	15.008	1724611	52470	570040.32
11)	ODK27	05/03/27	14.493	14.466	0.027			15.008	123495	1003	576705.72
12)	ODM27	06/01/27	14.528	14.512	0.016	06/03/27	0.20	15.010	48756	481	583141.31
13)	ODN27	07/01/27	14.545	14.555	-0.010	07/22/27	-1.48	14.995	2088573	54111	590304.26
14)	ODQ27	08/02/27	14.604	14.590	0.014			14.995	67538	209	597553.16
						09/02/27	-2.36	14.971			
15)	ODV27	10/01/27	14.640	14.643	-0.003	10/21/27	-1.79	14.953	1306526	2937	611960.36
16)	ODX27	11/01/27	14.675	14.662	0.013			14.953	35866	44	618771.17
17)	ODZ27	12/01/27	14.681	14.678	0.003	12/02/27	-2.18	14.931	42462	69	625652.75

Mexico

- Swap curve --- implied pricing hidden in swap curve
- 20 day horizon
- If you can call a fwd for next meeting, what is curr forward starting in day after that meeting
  - If higher or lower, take difference

Spot --> S(0,T)

Spot(today, 1m)

Fwd --> f(0,t,T)

- Needs expiration

Looking for 1m fwds starting from CB date

2y rate

1yr rate

1y1y ---> (1+f2)^2 = (1+f1)^1 (1+f1,1)^1 No arbitrage condition

- **Option to change curve source**
- CRVF (curve fitting, pick country, then pick curve)
  - Table for tickers

LATAM :

Assumptions for **Colombia**

0) Country
   a.
1) Curve Source
   a. OIS
2) Policy anchor
   a.
3) Valuation date (as of today)
   a.
4) Forward window (if meetings are monthly or not (ex: 28 days after each meeting)
   a. Meet 8 times a year
5) Day Count
   a. ACT/365
6) Interpolation (pchip)
   a.
7) Standard move (how many bps)
   a.
8) Basis
   a.
9) Scenario preset
   a.
10) Fair Value Method
   a.

Assumptions for **Chile**

11) Country
   a. CHILE
12) Curve Source
   a.
13) Policy anchor
   a.
14) Valuation date (as of today)

    a.

15) Forward window (if meetings are monthly or not (ex: 28 days after each meeting)
16) Day Count
    a. ACT/360
17) Interpolation (pchip)
    a.
18) Standard move (how many bps)
    a.
19) Basis
    a.
20) Scenario preset
    a.
21) Fair Value Method
    a.